

Top 10 Products

Ranking and Sorting: Retail Sales Dataset, December 2009 – December 2010

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Notebook	Top10_Products.ipynb	Tools	Python (Pandas), Excel-ready outputs

KEY RESULTS

- Of **4,251** distinct stock codes, **20** are not real products (postage, fees, manual entries, test rows, gift vouchers) and were excluded, removing **467,115.42** in non-product value.
- Without that cleanup, “**Manual**” would rank **#1** in a naive top 10, ahead of every real product, and **3** of the naive top 10 slots would be non-products.
- The top 10 real products bring in **772,683.54**, **7.9%** of all **4,231** products' sales.
- No product ties exactly at the rank-10 cutoff (39,762.78), so the top 10 is unambiguous, but **10** products do tie further down the list, confirming the tie-breaking rule was worth setting.

1. Project Brief

Task	Identify the top 10 products by sales.
Objective	Practice ranking and sorting.
Deliverables	Top-10 table and a chart.
Approach	Sort descending. Check ties.

2. Data and Method

The data is the cleaned copy from the Duplicate Record Check (6,865 exact duplicate rows removed), restricted to **sale lines**: rows with Quantity × Price above zero, on a normal invoice (not a cancellation starting with C) — the same rule used in the Basic Sales Summary and Monthly Sales Trend tasks, so the figures agree across all four notebooks. That leaves **504,730** sale lines across **4,251** distinct stock codes.

Step	What it does
1. Filter to sale lines	Same rule as the earlier KPI and Monthly Trend tasks: real sales only.
2. Check what a “StockCode” actually is	Most codes are 5 digits (optionally +1 letter). Anything else is inspected by hand.
3. Exclude non-products	20 codes removed (Section 3): postage, fees, manual entries, test rows, gift vouchers.
4. Group by product	Sum Sales and Quantity, count distinct Orders, per StockCode. Most common Description used where spelling varies.
5. Sort and rank	Sort by Total Sales, descending. Rank with ties handled explicitly (Section 5).
6. Take the top 10	The 10 highest-ranked real products (Section 6).

3. A Trap: Not Every Code Is a Product

43 of the 4,251 stock codes do not match the standard 5-digit pattern. Most of these (23) are still real products with an unusual code (party bags, dog collars, and similar items) and were kept. But 20 are not products at all:

Code	What it is	Rows	Sales value
M	Manual	550	262,577.36
DOT	DOTCOM POSTAGE	730	116,408.71
POST	POSTAGE	759	49,477.54
AMAZONFEE	AMAZON FEE	2	20,248.04
ADJUST	Adjustment by john on 26/01/2010 16	36	8,897.93
C2	CARRIAGE	131	6,625.00
ADJUST2	Adjustment by Peter on Jun 25 2010	3	731.05
gift_0001_30	Dotcomgiftshop Gift Voucher £30.00	17	434.56
+ 12 more small codes	Discounts, samples, test entries, gift vouchers		1,715.23

WHY THIS MATTERS

Left uncleaned, the naive ranking puts **“Manual” in 1st place** (a bookkeeping code, not a sale), with postage codes taking two more of the top 10 spots. A “Top 10 Products” list that is actually topped by “Manual” and “Postage” would mislead anyone using it to plan stock or marketing.

4. Adding Up Sales by Product

The remaining 502,430 sale lines were grouped by StockCode, giving 4,231 distinct real products. For each: Total Sales and Total Quantity are summed, and Orders counts the distinct invoices that included it. Where the same code had inconsistent spelling in Description (extra spaces, a typo), the most common spelling was used.

5. Ranking and Ties

Sorting orders the rows; ranking numbers them, and ranking is where ties become visible. Pandas offers three tie-breaking rules: **min** (tied items share the better rank, the next rank is skipped — like Olympic medals), **dense** (tied items share a rank, nothing is skipped) and **first** (ties broken by row order, hiding that a tie exists). This project uses **min**.

Method	Two products tied at the top	Next distinct value gets rank
min (used here)	Both get rank 1	3 (rank 2 is skipped)
dense	Both get rank 1	2 (nothing skipped)
first	Ranks 1 and 2 (tie hidden)	3

Checked, not assumed: the 10th-place cutoff is 39,762.78, and exactly 1 product sits at that value, so the top 10 here is unambiguous. Ties do exist further down the list — for example 10 products are tied at a sales total of only 2.55 each, mostly very small items. This confirms the rule matters in general, even though it made no difference to this particular top 10.

6. The Top 10 Products

Rank	Product	Total Sales	Total Qty	Orders	% of grand total
1	REGENCY CAKESTAND 3 TIER	169,912.76	13,685	2,019	1.7%
2	WHITE HANGING HEART T-LIGHT HOLDER	158,305.72	58,386	3,281	1.6%
3	JUMBO BAG RED RETROSPOT	88,976.33	49,804	1,966	0.9%
4	ASSORTED COLOUR BIRD ORNAMENT	72,890.19	45,228	1,412	0.7%
5	PAPER CHAIN KIT 50'S CHRISTMAS	58,127.30	17,205	957	0.6%
6	PARTY BUNTING	49,664.12	10,079	1,013	0.5%
7	ROTATING SILVER ANGELS T-LIGHT HLD	47,954.49	23,037	347	0.5%
8	RED RETROSPOT CAKE STAND	44,994.25	4,251	1,025	0.5%
9	DOOR MAT UNION FLAG	42,095.60	6,406	1,072	0.4%
10	DOOR MAT RED SPOT	39,762.78	5,880	1,066	0.4%

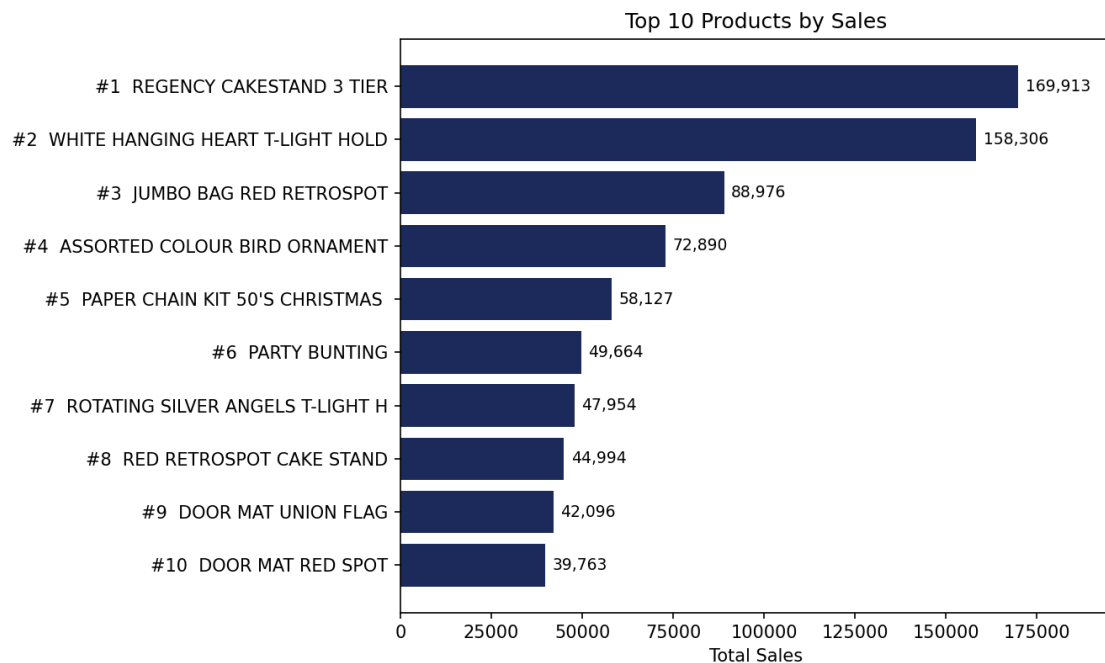


Figure 1: Top 10 products by sales. A horizontal bar chart is the standard choice for ranking named items — easy to compare, and long product names stay readable.

7. How Concentrated Are Sales?

Measure	Value
Total products (after excluding non-products)	4,231
Total sales across all products	9,804,647.24
Sales from just the top 10 products	772,683.54
Share of all product sales from the top 10	7.9%

Ten products, out of more than 4,000, bring in about 7.9 in every 100 of all product sales. This kind of concentration is common in retail (a small share of products driving a large share of revenue) and is exactly why a Top-10 view is a useful lens: it turns a list of over four thousand products into ten worth acting on.

8. Assumptions and Limitations

- **Non-product exclusion list is a judgement call.** The 20 codes excluded here were identified by inspecting every non-standard code by hand. A different analyst might classify a borderline case (e.g. gift vouchers) differently; the full list is documented in the notebook and in the Excel file's Method Notes sheet.
- **Ranking is by Total Sales value,** not by quantity sold or number of orders. A product sold in huge volume at a low price could rank highly by quantity but not appear here.
- **Ties are broken with method="min".** A different convention (dense or first) would not change which 10 products appear here, since no tie falls at the cutoff, but could change results on a different cut of the data.
- **Description spelling.** Some stock codes have more than one spelling of their description; the most common spelling was used, which is a simplification.
- **Scope.** Covers the whole period as one ranking. It is not split by month or country.

9. Interview Questions

How do you handle ties?

First decide whether a tie is even meaningful for the measure in question, then pick a rule and apply it consistently. Pandas' `rank()` offers **min** (tied items share the better rank, the next rank is skipped), **dense** (tied items share a rank, nothing skipped) or **first** (ties broken by row order). This project used min, and explicitly checked whether a tie fell at the rank-10 cutoff (it did not: only 1 product sits at 39,762.78) rather than assuming there wasn't one.

Why use Top-10 analysis?

It turns a long list (4,231 products here) into a short, actionable one: which products to keep well stocked, feature in marketing, or review with a supplier. It works because retail sales are usually concentrated in a few items — here the top 10 alone made up 7.9% of all product sales, so acting on a short list captures a disproportionate share of the business.

10. Conclusion and Recommendations

The top 10 products by sales were identified after a necessary cleanup step: excluding 20 non-product codes worth 467,115.42 that would otherwise have contaminated the ranking, including putting a bookkeeping entry in 1st place. The final top 10 are led by the **REGENCY CAKESTAND 3 TIER** (169,912.76) and together make up 7.9% of all product sales.

Recommendations

- Keep the non-product exclusion list under review as new data arrives; new administrative codes can appear.
- Consider a second Top-10 view ranked by Quantity sold, since it can surface high-volume, low-price products that do not appear in a sales-value ranking.
- Prioritise stock availability and marketing attention on the top 10, given the share of revenue they represent.
- Standardise product descriptions per StockCode at the source, so no analysis step has to guess the "correct" spelling.
- Re-run this ranking monthly or quarterly to catch new best-sellers or fading products early.

Appendix: Core Code

```
is_sale_line = (~Invoice.str.startswith("C")) & (Quantity * Price > 0)
sales = df[is_sale_line]
sales = sales[~sales.StockCode.isin(NON_PRODUCT_CODES)]

by_product = sales.groupby("StockCode").agg(Total_Sales=("Sales", "sum"), ...)
by_product = by_product.sort_values("Total_Sales", ascending=False)
by_product["Rank"] = by_product["Total_Sales"].rank(method="min", ascending=False)
top10 = by_product.head(10)
```